

# Lunar Greenhouse

Campaign 1 · Case 02 · Shackleton Crater, Lunar South Pole



SOLAR  
AGRICULTURAL  
AGENCY  
Student Mission

MISSION QUESTION

Why do healthy lunar tomato plants form flowers but no fruit?

YOUR ROLE

**Process Modeler** · Reconstruct the sequence, locate the first failure, then test the diagnosis.

SCIENCE FOCUS · ESTABLISHED + FICTIONAL CONTEXT

Tomato reproduction is a sequence: pollen must be released, reach a receptive stigma, germinate, grow a pollen tube, and complete fertilization before fruit set. The lunar greenhouse and exact readings are fictional case evidence.

 TASK 01 1 · Vocabulary

|               |                                               |
|---------------|-----------------------------------------------|
| Anther        | Structure that produces and holds pollen.     |
| Fertilization | Joining of reproductive cells at an ovule.    |
| Fruit set     | Beginning of fruit development.               |
| Pollen tube   | Growth from a pollen grain through the style. |
| Pollination   | Transfer of pollen to a receptive stigma.     |
| Stigma        | Receptive surface where pollen must arrive.   |

 TASK 02 2 · Initial thinking

A plant is vigorous and flowering but produces no fruit. Which reproductive step would you investigate first, and what evidence would you seek?

 TASK 03 3 · Model the pollination sequence

Use each word-bank phrase once. Place the six phrases in process order.

WORD BANK

fruit set · physical agitation · viable pollen in anthers · fertilization · pollen tube growth · pollen reaches stigma

1

PROCESS STEP

→

2

PROCESS STEP

→

3

PROCESS STEP

→

4

PROCESS STEP

→

5

PROCESS STEP

→

6

PROCESS STEP

PROCESS MODEL

Curriculum-original sequence schematic. Read left to right; arrows show dependency.

**Model rule:** A failed event can interrupt every later event. Do not assume a later structure is defective merely because the process never reaches it.

 TASK 04 4 · Identify the failed step

Mark the **first** step that fails. Use two observations to locate it and state one downstream result.

|               |                   |
|---------------|-------------------|
| Failed step   | Observation 1     |
| <div></div>   | <div></div>       |
| Observation 2 | Downstream result |
| <div></div>   | <div></div>       |

 TASK 05 5 · Test the competing explanations

Mark **Supported** or **Weakened**, then give one evidence-based reason.

| Possible diagnosis                                              | Classification | Evidence-based reason |
|-----------------------------------------------------------------|----------------|-----------------------|
| Lunar regolith is toxic to fruit development.                   | <div></div>    | <div></div>           |
| The light spectrum lacks ultraviolet light needed for fruiting. | <div></div>    | <div></div>           |
| No effective pollen release and transfer is occurring.          | <div></div>    | <div></div>           |
| Carbon dioxide is too high for tomatoes.                        | <div></div>    | <div></div>           |

 TASK 06 6 · Diagnose and reject an alternative

Diagnosis: state the cause and the failed biological step.

Rejected alternative: name it and give the evidence that weakens it.

 TASK 07 7 · Claim-Evidence-Reasoning

|                                                    |
|----------------------------------------------------|
| Claim                                              |
| 1–2 sentences                                      |
| <div></div>                                        |
| Evidence                                           |
| 2–3 specific pieces from more than one source      |
| <div></div>                                        |
| Reasoning                                          |
| Explain the mechanism connecting evidence to claim |
| <div></div>                                        |

 TASK 08 8 · Design a reliable pollination support

Propose a method that supplies the missing physical action without animal pollinators.

|                |                                                     |
|----------------|-----------------------------------------------------|
| Design         | One criterion                                       |
| <div></div>    | <div></div>                                         |
| One constraint | Mechanism check: restored step and success evidence |
| <div></div>    | <div></div>                                         |

 TASK 09 9 · Exit ticket

A second sealed habitat grows a different self-pollinating crop. Flowers and viable pollen are present, but no fruit forms. Which step should investigators test first, and what observation would show whether it is failing?